

Problem 2 : $v_p(t) = m(t) \sin(2\pi f_c t + \phi_c)$

$$\begin{aligned}
 1) \quad y(t) &= x(t) \sin(2\pi f_c t + \phi_r) \\
 &= v_p(t-d) \sin(2\pi f_c t + \phi_r) \\
 &= m(t-d) \sin(2\pi f_c(t-d) + \phi_c) \sin(2\pi f_c t + \phi_r) \\
 &= \frac{m(t-d)}{2} \left(\cos(4\pi f_c t - 2\pi f_c d + \phi_c + \phi_r) + \cos(2\pi f_c d + (\phi_c - \phi_r)) \right)
 \end{aligned}$$

2. Cut-off: $B < f < f_c \cdot H_z$.

$$v(t) = \cos(2\pi f_c d + (\phi_c - \phi_r)) \cdot \frac{m(t-d)}{2}$$

3. $d = n T_c + T_c/4$

$$\begin{aligned}
 \phi_c = \phi_r \Rightarrow v(t) &= \cos(2\pi f_c n T_c + 2\pi f_c \cdot \frac{T_c}{4}) \\
 &= \cos\left(2\pi n + \frac{\pi}{2}\right) \cdot \frac{m(t-d)}{2}
 \end{aligned}$$

$$\begin{aligned}
 \text{for } \frac{m(t-d)}{2} \left\{ \begin{array}{l} \text{max value} = \frac{\max(m(t-d))}{2} \\ \text{min value} = \frac{\min(m(t-d))}{2} \end{array} \right. , \quad \cos\left(2\pi n + \frac{\pi}{2}\right) = \cos\left(\frac{\pi}{2}\right) = 0 \\
 \Rightarrow \max(v(t)) = \min(v(t)) = 0.
 \end{aligned}$$